

SETUP & INSTALLATION GUIDE

TinyFish.ai + Hermes Agent Complete Setup Guide

How to Auto-Enrich Any Spreadsheet with Live Web Data — Free

Based on the video: "How to Use TinyFish to Auto-Enrich Any Spreadsheet (FREE)" • May 2026

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1. Overview: What You Are Building

This guide covers the complete end-to-end setup for a workflow that takes a blank spreadsheet and automatically fills it with live web data — no manual copy-pasting, no browser tabs, no API wrappers to build.

The stack has two components working together:

- **TinyFish.ai** — web infrastructure for AI agents. It provides a real browser fleet so that agents can search the live web, fetch clean page content, and navigate JavaScript-heavy sites. ¹
- **Hermes Agent** — an open-source, self-improving AI agent by Nous Research. It reads your spreadsheet, decides what to search for per row, fires the searches in parallel via TinyFish, and writes results back into the file. ²

Two TinyFish skills — `use-tinyfish` and `enrich-excel` — are installed into Hermes to give it these capabilities. Both are free and open-source, published in the TinyFish Cookbook on GitHub. ³

Time to complete this setup: Under 30 minutes if you follow this guide step by step. TinyFish Search and Fetch are free with no credit card required. Hermes is MIT-licensed. Railway infrastructure starts at approximately \$5/month.

Prerequisites Checklist

- A GitHub account (for Railway deployment)
- An email address to register for TinyFish
- An LLM API key (OpenRouter is the easiest starting point — free sign-up, some free models)
- Optionally: a Telegram account (fastest messaging channel to connect)

2. Part 1 — TinyFish.ai: Setup & API Key

TinyFish gives AI agents the ability to use the real web. The platform runs its own Chromium fleet with stealth handling and full JavaScript rendering. As of May 4, 2026, the Search and Fetch endpoints are completely free for all users with no billing setup required. ⁴

Step 1: Create Your Account and Get an API Key

- 1 Go to **agent.tinyfish.ai** and sign up for a free account. No credit card is required. ⁴
- 2 After signing in, navigate to **agent.tinyfish.ai/api-keys** and click **Create Key**. Copy the generated key and store it securely — you will not be able to view it again.

Free Tier Rate Limits

The free tier has the following limits as of May 2026. ⁴

API	Cost	Rate Limit	Endpoint
Search	FREE	5 queries per minute	<code>api.search.tinyfish.ai</code>
Fetch	FREE	25 URLs per minute	<code>api.fetch.tinyfish.ai</code>
Browser	PAID (CREDITS)	Depends on plan	<code>api.browser.tinyfish.ai</code>
Agent	PAID (CREDITS)	Depends on plan	<code>agent.tinyfish.ai/v1/automation/...</code>

For the spreadsheet enrichment workflow in this guide, only **Search** is required for most rows. Fetch is the escalation path when a pricing page needs full content extraction. Both are free.

3. Part 2 — The Four TinyFish APIs Explained

TinyFish exposes four API surfaces under a single API key. Understanding the escalation ladder is essential because the `use-tinyfish` skill teaches Hermes to start with the lightest tool and escalate only when needed. ¹

Tool	Use When	Speed	Cost
Search	You need to find URLs or get a quick answer about a topic. Returns ranked results with titles, URLs, and snippets.	Fastest	FREE
Fetch	You already have URLs and need their full, clean content — strips ads, nav bars, cookie banners. Returns Markdown, JSON, or HTML.	Fast	FREE
Browser	You need to interact with a page: click buttons, fill forms, navigate dynamically. Also used for login-gated sites (pass your credentials via Vault).	Slower	CREDITS
Agent	Full browser automation via natural-language goals. The agent opens a real browser, navigates, clicks, and extracts structured data — all described in plain English.	Slowest	CREDITS

Common Escalation Patterns

The `use-tinyfish` skill teaches Hermes four standard escalation patterns used in practice: ⁶

- **Research:** `search` to find URLs, then `fetch` the best results to read full content.
- **Deep extraction:** `search` to find the right site, then `agent` to interact and extract structured data.
- **Escalation:** `fetch` first; if the page is dynamic or returns empty content, escalate to `agent`.
- **Full control:** when `agent` cannot handle a complex multi-step workflow, spin up a raw `browser` session for direct CDP control.

MCP Integration (for Claude, Cursor, Codex)

TinyFish also exposes all four tools via a Model Context Protocol (MCP) server. Drop the server URL into any MCP-aware client — Claude, Cursor, Codex, ChatGPT desktop — and those tools get

TinyFish access without any extra CLI setup. ¹ This is an alternative integration path but not required for the Hermes skills workflow covered in this guide.

n8n Integration

TinyFish publishes pre-built n8n workflow nodes for users who prefer visual automation. Import the JSON files from the TinyFish Cookbook's `N8N_WorkFlows` folder and add your API key as a credential. ³

4. Part 3 — Hermes Agent: Deploy on Railway

Railway is a cloud platform that runs Hermes as a persistent service with a web admin dashboard, managed TLS, and persistent storage. This is the path recommended in the video — no Docker configuration, no machine to leave running, no reverse proxy setup. ⁸

Step 1: Deploy via the Railway Template

- 1 Go to railway.com/deploy/hermes-agent-by-no-1 and click **Deploy Now**. ⁸
- 2 Sign in with your GitHub account. Railway will provision a project automatically.
- 3 You will be prompted to set two environment variables:

Variable	Description	Required
<code>ADMIN_USERNAME</code>	Username for the admin dashboard (default: <code>admin</code>)	No
<code>ADMIN_PASSWORD</code>	Password for the admin dashboard. If left blank, a random password is generated and printed to deploy logs.	No

All other configuration — LLM provider keys, model selection, channel tokens — is handled through the admin dashboard, not environment variables.

- 4 Wait for the deployment to complete (typically 2–4 minutes). Railway will provision a public URL for your instance.

Step 2: Access the Admin Dashboard

- 5 Open your Railway app URL in a browser. Log in with the `ADMIN_USERNAME` and `ADMIN_PASSWORD` you set. If you did not set a password, check the Railway deployment logs — a random password will be printed there. ⁸

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On the **Setup** page, select your LLM provider from the dropdown and configure DeepSeek:

1. Sign up at **deepseek.com** and navigate to the API section.
2. Create an API key and copy it.
3. In the Hermes dashboard, enter `deepseek-v4-pro` as the model.
4. Paste the API key into the dashboard field and save.

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Enable at least one messaging channel. **Telegram is the fastest:**

1. Message @BotFather on Telegram, send `/newbot` , and copy the token.
2. Paste the token into the Telegram channel field in the dashboard.

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Click **Save & Start**. Message your Telegram bot. Approve the pairing request that appears in the dashboard under the **Users** tab.

Cost: Railway infrastructure starts at approximately \$5/month. LLM costs depend on your provider — OpenRouter offers free-tier models, making the full stack runnable at zero LLM cost during testing.

5. Part 4 — Installing the TinyFish Skills into Hermes

Skills are reusable instruction sets that extend Hermes with new capabilities. You install a skill once, and Hermes knows how to use it from that point on. TinyFish publishes two skills in their open-source Cookbook that are required for the spreadsheet enrichment workflow. ³

The Two Required Skills

Skill	What It Does	GitHub URL
<code>use-tinyfish</code>	Teaches Hermes the four TinyFish tools (Search, Fetch, Browser, Agent), when to use each one, and how to call them. Defines the escalation ladder and handles authentication.	skills/use-tinyfish
<code>enrich-excel</code>	Teaches Hermes to read a spreadsheet's headers, identify empty cells, generate targeted search queries per row, fire them in parallel via TinyFish, and write results back into the file.	skills/ (enrich-excel folder)

How to Install: Just Type It in Chat

There is no CLI command, no file cloning, and no dashboard configuration required. Hermes installs skills directly from a GitHub URL when you tell it to in plain English. Open your Hermes chat — via Telegram, Discord, or the web dashboard — and send these two messages, one at a time:

- 1 Install the core TinyFish tool skill:

```
Install this skill https://github.com/tinyfish-io/tinyfish-cookbook/tree/main/skills/use-tinyfish
```

Hermes will fetch the skill file from GitHub, read the instructions, and confirm the installation. This gives Hermes knowledge of the four TinyFish tools and when to use each one.

2

Install the spreadsheet enrichment skill:

```
Install this skill https://github.com/tinyfish-io/tinyfish-cookbook/tree/main/skills
```

Hermes will fetch the skills folder from GitHub and install the `enrich-excel` skill (and any other available skills in that folder). This gives Hermes the ability to read a spreadsheet, generate per-row queries, run them in parallel, and write results back.

That is the entire installation. No cloning, no file paths, no slash commands, no dashboard tabs. Two sentences typed into the chat. Hermes handles fetching, parsing, and activating each skill automatically.

Verify the Skills Are Active

After both installs, confirm Hermes recognises the new capabilities by asking:

```
What TinyFish tools do you have available?
```

Hermes should describe the four TinyFish tools — Search, Fetch, Browser, and Agent — and explain the escalation order. If it does not, re-send the install message for the skill that is missing.

What Each Skill Teaches Hermes

Once installed, the `use-tinyfish` skill gives Hermes four web tools in escalation order: ⁶

Tool	What Hermes Does With It	When It Is Used
Search	Runs a web search and returns ranked results with titles, URLs, and snippets	First attempt for every row — fastest, free, no credits
Fetch	Renders a URL in a real browser, strips noise, and returns clean Markdown	Escalation when a Search snippet lacks enough detail
Browser	Opens a headless browser for interaction: click, scroll, fill forms	Login-gated pages or multi-step navigation (uses credits)
Agent	Full natural-language browser automation — describe a goal, agent executes it	Complex extraction where Browse alone is insufficient (uses credits)

For the spreadsheet enrichment workflow, Hermes almost always stays within the free Search and Fetch tools. The paid Browser and Agent modes are only triggered if a pricing page is behind a login or requires navigation to reveal the data.

6. Part 5 — Running Your First Spreadsheet Enrichment

With TinyFish and both skills installed, you can now enrich any spreadsheet with live web data using a single prompt.

Step 1: Prepare Your Spreadsheet

Create an Excel (.xlsx) or CSV file with:

- **Column headers** that describe what data you want (e.g. "Free Tier", "Starter Plan Price", "Credits Included", "Model Access").
- **Row data** in the first column identifying each item to research (e.g. company names, product names, addresses).
- **Empty cells** in all other columns — these will be filled by Hermes.

Tip: The more descriptive your column headers, the better the search queries Hermes generates. "Starter Plan Monthly Price USD" produces more accurate results than just "Price".

Step 2: Upload the File to Hermes

How you upload depends on your deployment:

- **Local install:** place the file in a directory accessible to Hermes and reference its path in your prompt.
- **Telegram:** attach the .xlsx file as a document message in your Telegram chat with the Hermes bot.
- **Railway web dashboard:** use the file upload feature in the dashboard's chat pane.

Step 3: Send the Enrichment Prompt

Send exactly one sentence. The `enrich-excel` skill handles everything else:

```
Enrich this competitive monitor using TinyFish. Research each coding agent's pricing plans and fill in the empty columns from their live websites. Parallelize the processing and make sure to verify data scraping results.
```

You can adapt this prompt for any domain:

- **Lead research:** "Enrich this company list using TinyFish. Find funding stage and employee count for each company."
- **Salary benchmarking:** "Enrich this job titles spreadsheet using TinyFish. Find current salary ranges from live job boards."
- **Vendor pricing:** "Enrich this software tools list using TinyFish. Find current pricing for each tool from their live websites."
- **Real estate comps:** "Enrich this address list using TinyFish. Find days on market and price per square foot."

Step 4: What Happens Inside Hermes

The `enrich-excel` skill orchestrates the following sequence automatically:

1. **Header analysis:** Hermes reads all column headers to understand what data is needed for each column.
2. **Query generation:** for each row, Hermes constructs a targeted search query by combining the row identifier with the column intent (e.g. "GitHub Copilot pricing free tier 2026").
3. **Parallel Search calls:** all row queries fire simultaneously via `tinyfish search` — not one at a time. This is the key performance advantage.
4. **Escalation if needed:** if a Search snippet does not contain enough detail, the skill escalates to `tinyfish fetch` to pull the full pricing page content.
5. **Result writing:** Hermes maps the extracted data to the correct column, row by row, and writes the completed file back to the same location.

Step 5: Review the Output

When Hermes finishes, it will deliver the enriched file and summarize what it found. Review carefully:

- Cells where pricing is only available behind a sales conversation may be left blank or noted as "Contact sales" — this is the correct behavior.
- Dynamic or JavaScript-heavy pricing pages may require the escalation to Fetch or Agent mode. If Search snippets are insufficient, Hermes handles this automatically when the skill is configured correctly.
- For verification, spot-check 2–3 rows against the live websites.

Example: 15-Row Coding Agent Competitive Monitor

The video demonstrates this exact workflow on a spreadsheet tracking 15 AI coding agents (GitHub Copilot, Cursor, Windsurf, Claude Code, and others) with columns for Free Tier, Starter Plan Price, Credits Included, and Model Access.

- Total time from sending the prompt to receiving the completed file: under 60 seconds.
- All 15 TinyFish Search calls fire in parallel.
- For most rows, Search snippets alone contain sufficient pricing data.
- Rows where pricing is gated behind enterprise sales conversations are flagged rather than guessed.

7. Part 6 — Troubleshooting & Known Limitations

Common Issues and Fixes

Symptom	Likely Cause	Fix
Hermes starts but gives empty or broken replies	LLM provider auth or model selection is wrong	Run <code>hermes model</code> and re-confirm provider, model, and API key. Run <code>hermes doctor</code> .
TinyFish search returns 401 / unauthorized	<code>TINYFISH_API_KEY</code> not set or set incorrectly	Run <code>echo \$TINYFISH_API_KEY</code> to confirm the key is exported. Re-export if missing.
Skills not recognized by Hermes after install	Skill file path was wrong, or session needs restart	Run <code>/skills list</code> in a new session. Reinstall via the correct path.
Hermes on Railway cannot reach TinyFish	<code>TINYFISH_API_KEY</code> not set in Railway environment variables	Add the variable in the Railway service Variables tab and redeploy.
Spreadsheet enrichment returns no data for a row	Pricing is behind a paywall or enterprise gate	Normal behavior — the skill marks these cells rather than guessing. Manually check those rows.
Gateway starts but Telegram messages do not arrive	Bot token invalid or user not approved	Re-run <code>hermes gateway setup</code> . Check Users tab in dashboard for pending approvals.
Rate limit errors from TinyFish Search	Free tier: 5 queries per minute exceeded	Reduce row count per batch or upgrade at tinyfish.ai/pricing for higher limits.
Model context error on startup	Selected model has less than 64K token context	Switch to a model with at least 64K context via <code>hermes model</code> .

Known Limitations

- **Publicly available pricing only:** the workflow works best when companies publish pricing on their websites. Tools with enterprise-only or "contact us" pricing will produce incomplete rows.

- **Accuracy depends on page structure:** if a pricing page is a JavaScript single-page app that does not render key content in the initial HTML, Hermes may need to escalate from Search to Fetch or Agent mode.
- **Rate limits on the free tier:** 5 Search queries per minute. For sheets with 15–20 rows firing in parallel, this can be hit. If you see rate limit errors, process in smaller batches or upgrade to a paid TinyFish plan.
- **Windows native support is early beta:** for the most stable Windows experience, use WSL2. ²
- **File upload via Telegram:** very large Excel files may need to be referenced by path rather than uploaded directly via Telegram.

Diagnostic Commands

```
# Run Hermes diagnostics
hermes doctor

# Check which skills are installed
# (inside a Hermes chat session)
/skills list

# Test TinyFish CLI directly
tinyfish search "test query"
tinyfish fetch content get --url https://example.com

# Check gateway status
hermes gateway status
```

Updating Hermes

To update Hermes, open your Railway project dashboard and select **Redeploy**. Railway will pull the latest version and restart the service automatically.

Resource Links

Resource	URL
TinyFish free API key	agent.tinyfish.ai/api-keys
TinyFish documentation	docs.tinyfish.ai
TinyFish Cookbook (skills & recipes)	github.com/tinyfish-io/tinyfish-cookbook
use-tinyfish skill file	tinyfish-cookbook / skills / use-tinyfish / SKILL.md
Hermes Agent GitHub	github.com/NousResearch/hermes-agent
Hermes documentation	hermes-agent.nousresearch.com/docs
Hermes Railway one-click deploy	railway.com/deploy/hermes-agent-by-no-1
OpenRouter (LLM provider)	openrouter.ai
TinyFish pricing (paid plans)	tinyfish.ai/pricing

References

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